

Swinging Device

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Designation

BA - Number

Technical Data

BA 911251-03-4

Safety

BA 911198-00-4

Structure and Function

BA 911199-03-4

Swinging Device

Technical Data

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1. General

The swinging device consists of

the machine "Hydraulic Cabinet HSAV"

and

the assembly group "Swinging Device cpl.".

2. Identification

Overall-Weight :¹⁾

Drawing :¹⁾

Parts List :¹⁾

Hydraulic Cabinet

Machine-Type : HSAV 125 / 50

Machine-No. :¹⁾

Machine-Weight :¹⁾

Machine-Drawing : see documentation of manufacturer

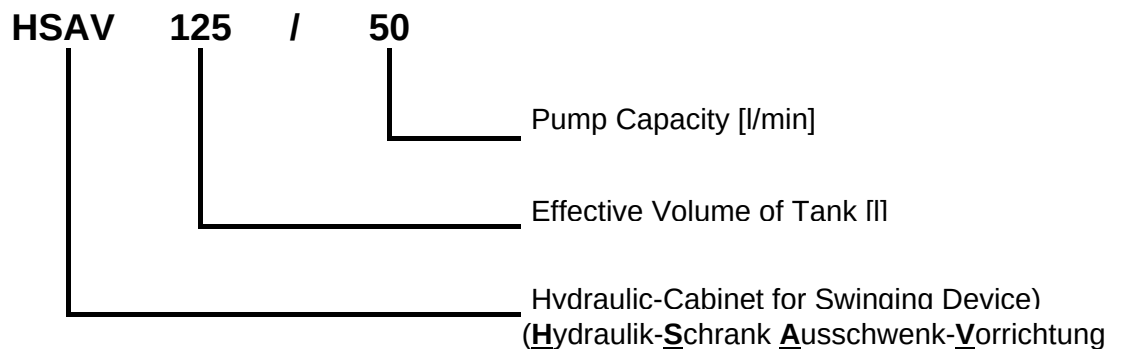
¹⁾ detailed information: see parts list of swinging device

Swinging Device cpl.

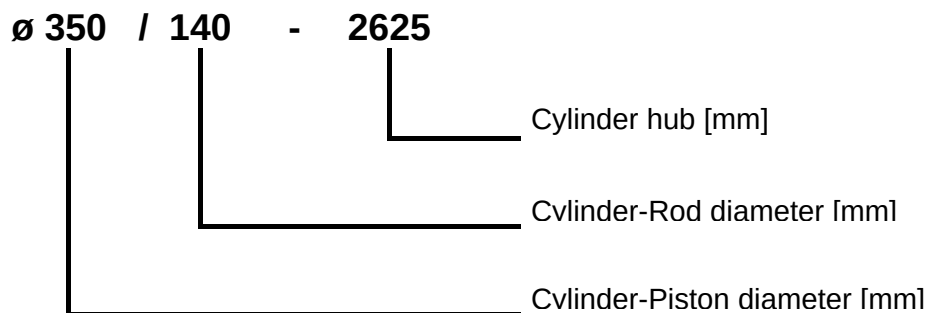
Assembly-Weight	:	 ¹⁾
Assembly-Number	:	1
Assembly-Drawing	:	 ¹⁾
Assembly-Parts List	:	 ¹⁾
Hydraulic Cylinder-Type	:	ø 350 / 140 - 2625
Hydraulic Cylinder-Drawing	:	see documentation of manufacturer

3. Machine-Type Designation

Hydraulic Cabinet



Hydraulic Cylinder



Swinging Device

Safety

1. General

These operating instructions swinging device - "Safety" is the supplement to the Roller Grinding Mill - "Safety" instructions. Ensure that these operating instructions will be studied in detail prior the application of the swinging device.

2. Notes on Safety at Work

- When performing any work on the mill in connection with the swinging in and out of the rollers, the staff must be protected accordingly. When platforms and railings are removed, the staff involved shall be safeguarded by belts.
- No persons shall be allowed to stay within roller swinging range when swinging the rollers in and out.
- When removing a complete roller from the rocker-arm, the latter must remain connected with the hydraulic swinging-out cylinder. Thus you avoid that the rocker-arm will also be lifted due to frictional contact in the tapered clamping sleeves as the roller axle is being raised and will snap back as soon as the axle becomes free. **Accident hazard!**
- Operation of the hydraulic cylinder without the pipe-burst safety devices is forbidden!
- Check high-pressure hoses for damage and replace them, if necessary, before use.
- Only high-pressure hoses rated for the appropriate pressure classification may be used.
- Quality, cleanliness and operating viscosity of the hydraulic medium are decisive for safety of operation, economy and service life of the machine.
- Use only the grade of hydraulic oil recommended by LOESCHE. The class of purity of the employed oil at least should conform to **NAS 1638 Class 8 or better.**

Swinging Device

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1. Purpose

The hydraulic swinging mechanism constitutes an auxiliary device (tool) enabling fast changing of the grinding parts in the LOESCHE-Mill. This facility, to a large extent, replaces both manpower and lifting appliances. Additionally, besides ease of servicing, accident danger will be diminished.

With the help of a hydraulic cylinder, this device permits the M- or S-roller to be swung out of the LM-upper housing. Moreover, it serves the purpose of supporting the roller held inside the rocker arm when, during swinging, the roller has gone beyond the instable point (culmination point of the centre of gravity).

In swung-out condition of the rocker arm/roller system, the roller axle is positioned vertically, with the piston rod in retracted position.

The swinging mechanism will likewise facilitate removal of the grinding plate segments on the table. If segment replacement work does not coincide with replacement of the roller tires, it will be sufficient to swing out one roller, the free body opening enabling then undisturbed manipulation with the lining segments.

2. Structure

The swinging device may be seen on the drawing "Swinging-Out Device cpl.". It depicts the swinging-cylinder (hydraulic cylinder) in its operational position. The supply system is a hydraulic cabinet with control elements mounted on a mobile carriage (HSAV).

The operating system consists principally of a double-acting hydraulic cylinder. The hydraulic cylinder itself has a joint which is slipped into a joint seating and secured against undue shifting. This joint enables swinging of the hydraulic cylinder during the swinging-out process. The piston rod of the hydraulic

cylinder is equipped with a fork element with bolt and split-pin retainer for attachment to the toggle joint of the rocker arm (with roller) to be swung out.

At its two ends, the hydraulic cylinder is provided with throttle-check valves. Quick release couplings enable the connection of high-pressure hoses. The couplings can be opened under pressure, without any loss of oil.

The supply system - the HSAV - is composed of a two-wheel carriage on which an oil tank and pump are mounted, a hand-operated control valve, an electric switch box, two high-pressure hoses for interconnection with the hydraulic swinging-cylinder and an electric cable for connecting the pump motor to the power supply.

With the help of the control lever of the valve, the operating direction of the hydraulic cylinder is preselected and the pump is set in action.

The pump runs as long as this control lever is being used. In the retracted position of the piston, the cylinder has to be filled with oil.

Topping-up of oil up to the tank cover is not permitted because the oil displaced out of the hydraulic cylinder while the piston rod is moved in, must be allowed to flow back into the tank. If there were not a sufficient volume reserve for this oil quantity inside the tank, it would unavoidably overflow.

The electric switch box is mounted on the tank of the HSAV and wired.

3. Preparatory work for Swinging-Out

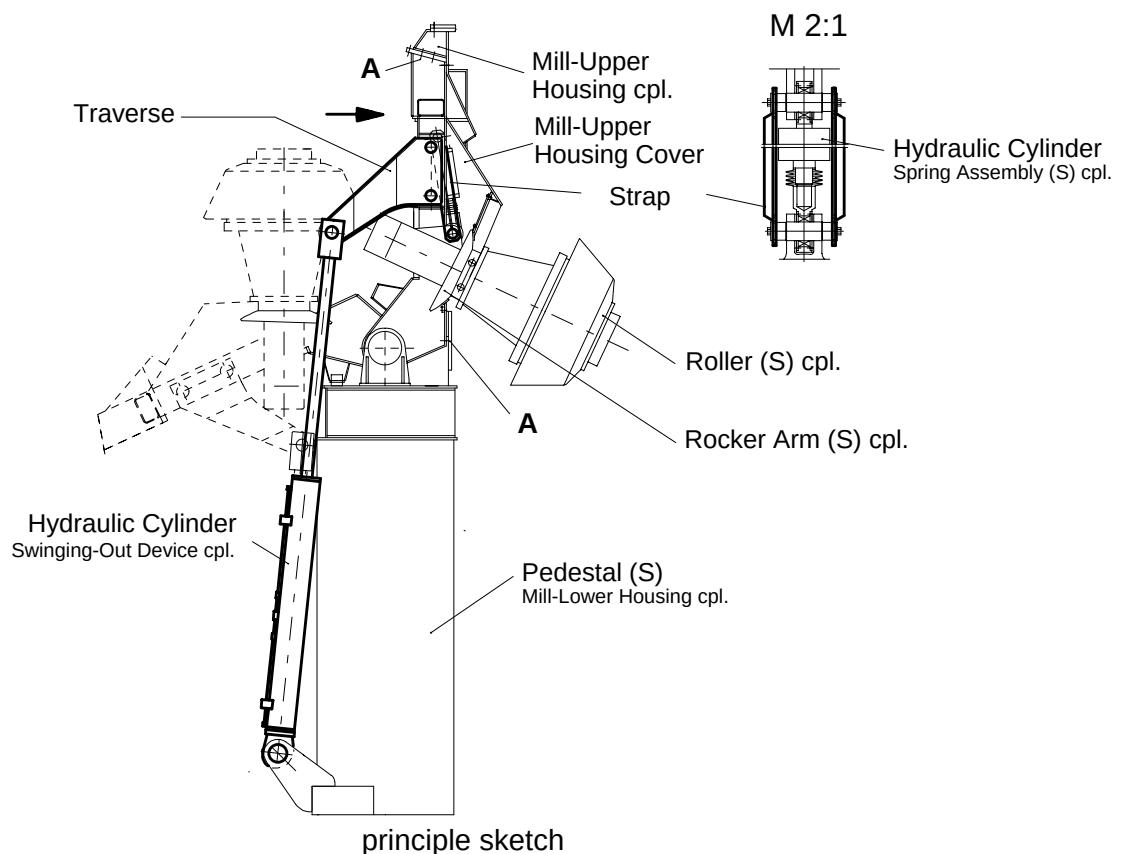
3.1 M-Roller

(please refer to design drawing "Rocker Arm (M) cpl.")

- a) Loosen taper pin connection between rocker arm and fork of the assembly "Rocker Arm (M) cpl." (see operating instructions "LM-Auxiliaries: M-Rocker-Arm", which is part of the LM-mill);
- b) Attach connecting strap between upper linkage point of the hydraulic cylinder and M-rocker arm;
- c) Unscrew cover of LM-upper housing and remove;
- d) Remove protection plates – if existing – of M - roller cpl.;
- e) Remove mill platform - if existing - away from the swinging area of the M-rocker arm;
- f) Provide a drain pan to collect any oil coming out of the M-roller (recirculating oil lubrication);
- g) Shut ventilation bore at the roller;
- h) Bring hydraulic cylinder to the desired position;

- i) Put hydraulic cylinder into the mounting bracket and secure cylinder in place;
- j) Bring HSAV towards hydraulic cylinder;
- k) Put the two hydraulic hoses of the HSAV with their quick-release couplings on to the corresponding cylinder connections;
- l) Connect the HSAV via the cable with power-supply plug;
- m) Move out piston rod by pushing the control lever of the valve into the correct position. This also activates the pump motor. At initial commissioning or after replacing the seals the piston should be driven in-and-out whilst the cylinder is bled of entrained air. Finally move piston rod out up to toggle joint of rocker arm;
- n) Interconnect fork element of piston rod, via bolt, with toggle joint of rocker arm;
- o) Insert split-pin for retaining bolt in corresponding bore.

3.2 S-Roller



- a) Attach traverse to cover of mill-upper housing (if present);
- b) Connect bolts of hydraulic cylinder of S-spring assembly with straps;
- c) Uncouple hydraulic hoses;
- d) Remove screw connections (A and lateral) between mill-upper housing and mill-upper housing cover;
- e) Connection and operating of HSAV and swinging-cylinder are similar to swinging-out process of M-Roller (from point g on).

4. Swinging-Out Process

- Set hand lever of control valve to position "Swinging-Out";
- The pump motor runs;
- With central positioning of control lever the pump motor will stop;
- Keep valve lever in working position until the axle of the swung-out roller is positioned vertically;

NOTE: When the lever is freed the valve automatically returns to position "0" through spring action. During swinging-out, the roller, together with the rocker arm, moves away from the table until its centre of gravity is precisely above the rocker arm axle bearing. Under the effect of further swinging-out, the weight of roller and rocker arm will assist the high-pressure pump. In order to avoid uncontrolled rapid motions, the oil flow is braked by the previously mentioned throttle-check valves in the hydraulic cylinder. These are adjusted to their minimum setting by the hydraulic cylinder manufacturer. Should a further adjustment be required care must be taken that the oil flowrate does not become so high that the safety valves integrated in the hydraulic cylinder are inadvertently activated!

- The rocker arm strokes downwards to its end position which is limited through a mechanical stop; a "softer" support can be achieved by fixing a wooden block to the mechanical stop before the swinging out procedure begins.
- Return valve lever to position "0". Pump motor stops.

NOTE: The hydraulic system is protected by an overflow valve forming an integral part of the pump unit.

The HSAV has a motor-protection switch with bimetallic tripping relay.

5. Swinging-In Process

The same process as described in chapter 4, will take place during swinging-in, however, in the reverse sequential order. In this case, the control valve has to be pushed to position "swinging-in".

The swinging-in operation is completed when the rocker arm has reached its normal working position.

6. Servicing

- Pins and mounting brackets should, at regular intervals, be checked for possible damage and rust defects;
- Hydraulic cylinder and piston rod have to be protected against external damage;
- Please store the high-pressure hoses in a safe place. When replacing them, hoses rated at a burst pressure of more than 200 bar (2900 psi) must be used;
- When removing a complete roller from the rocker arm, the latter has to remain coupled with the hydraulic cylinder. Otherwise there would be the danger of the rocker arm being lifted as well during lifting of the roller axle on account of friction in the conical clamping sleeves and the striking back as soon as the axle is free (accident danger!);
- Store hydraulic cylinder vertically after use. With horizontal storage piston and piston rod seals might become permanently deformed under the influence of the piston weight, so the during re-use, the necessary operating pressure will not be attained again due to leakage!